Graphical Abstract

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Highlights

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- Research highlights item 1
- Research highlights item 2
- Research highlights item 3

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ABSTRACT

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1. Introduction

2. Numerical methodology

2.1. Overall numerical framework

A two-dimensional steady aero–hydraulic framework was developed to investigate the aerodynamic behaviour of airfoils containing passive internal pores. The external flow is modelled using an incompressible, inviscid and irrotational source–vortex panel method, while the pressure-driven flow through each internal passage is evaluated using the Hagen–Poiseuille and Darcy–Weisbach relations, selected according to the Reynolds number of the flow within the circular pore. The aerodynamic and hydraulic models are coupled through the pressure and surface-normal velocity at the pore openings.

For each pore, the external aerodynamic solution provides the pressure at both ends of the pore. The resulting pressure difference drives flow through the internal passage from the higher-pressure end towards the lower-pressure end. This internal flow introduces suction at one end and blowing at the other, which are represented in the panel method through surface-normal transpiration velocities. The modified boundary condition then changes the external velocity and pressure fields, which in turn affect the pressure difference driving the pore flow. The external and internal flows are therefore solved as a coupled aero–hydraulic system.

The calculation begins from the corresponding impermeable (solid) airfoil solution, with the surface-normal transpiration velocity initially set to zero. The resulting surface pressures provide the initial pressure differences across the pores and hence the initial internal flow rates. The aerodynamic solution, pore pressure differences and transpiration velocities are then updated iteratively until a consistent coupled solution is obtained. This procedure is applied independently to each of the investigated pore heights, $H = 8$, 9 and 10 mm, and the converged aerodynamic and internal-flow quantities are retained for analysis.

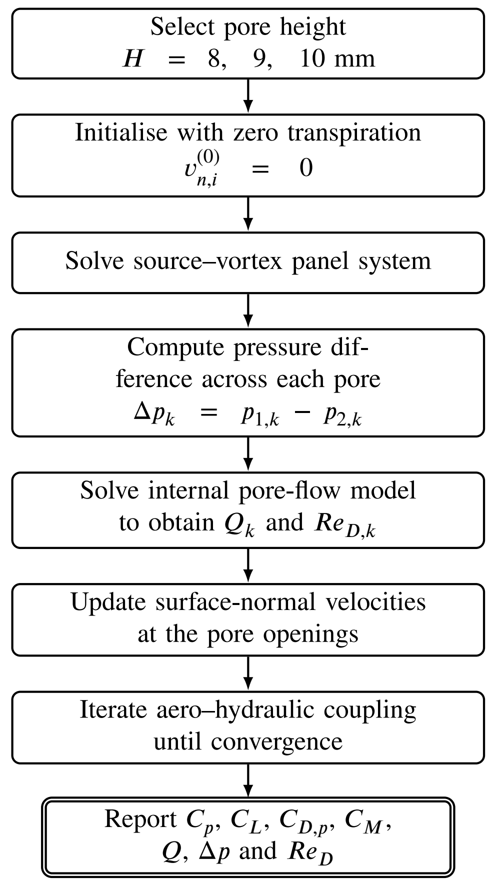

Figure 1: Coupled aero–hydraulic solution procedure applied to the investigated pore heights.

The aerodynamic model assumes steady, incompressible, inviscid and irrotational flow. It therefore does not resolve boundary-layer development, skin-friction drag, flow separation or stall. Consequently, the predicted drag coefficient represents only the pressure-drag contribution obtained from the potential-flow solution.

The numerical procedure is summarised in Fig. 1.

2.2. Airfoil geometry and porous configurations

The baseline geometry is a NACA 0018 airfoil with chord c. The NACA four-digit coordinates are generated analytically and discretised using cosine spacing along the

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chord, providing increased panel density near the leading and trailing edges.

Each aerodynamic panel i is characterised by its length S_i, control point (x_i, y_i), unit tangent vector

$$\boldsymbol{t}_i = (t_{x,i}, t_{y,i}), \tag{1}$$

and outward unit normal vector

$$\boldsymbol{n}_i = (n_{x,i}, n_{y,i}). \tag{2}$$

The porous structure is represented by passive circular channels connecting pairs of openings on the airfoil surface. No prescribed mass-flow rate or external pumping mechanism is imposed. Instead, the magnitude and direction of the internal flow are determined entirely by the aerodynamic pressure difference between the two surface openings.

Three porous configurations are considered:

1. **Horizontal model (M1):** four independent chordwise passages, with two channels located above and two below the chord line. Both openings of each passage are located on the same airfoil surface.
2. **Vertical model (M2):** four independent lower-to-upper passages distributed between $x/c = 0.1$ and $x/c = 0.9$.
3. **Combined independent model (M3):** the horizontal and vertical passages are superposed, resulting in eight internal channels. Although the channels may cross geometrically in the two-dimensional representation, they do not physically intersect or form internal junctions. Each passage remains hydraulically independent, and no flow is exchanged between the horizontal and vertical channels.

The three porous-airfoil configurations considered in the present study are illustrated in Fig. 2.

The positions of the surface openings are obtained by interpolation of the upper and lower airfoil surfaces rather than by restricting their locations to panel control points. This preserves the prescribed orientation and position of the internal passages independently of the aerodynamic panel resolution.

The channel geometry is checked before the coupled calculation to ensure that the openings remain within the permitted chordwise region, do not overlap, fit within the local airfoil thickness and satisfy the specified minimum separation.

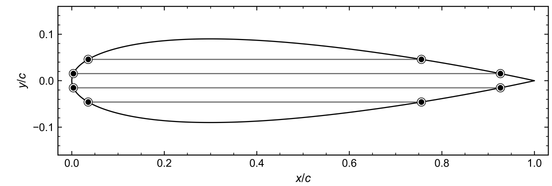
Each passage k is characterised by a diameter D_k and a centreline length L_k. The channel length is determined from the geometry of the two corresponding surface openings and the internal passage connecting them.

2.3. External aerodynamic model

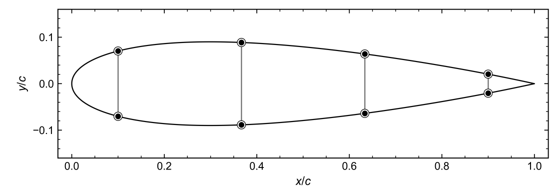
2.3.1. *Potential-flow formulation*

The external flow is assumed to be incompressible, inviscid and irrotational. The velocity field can therefore be expressed in terms of a scalar velocity potential,

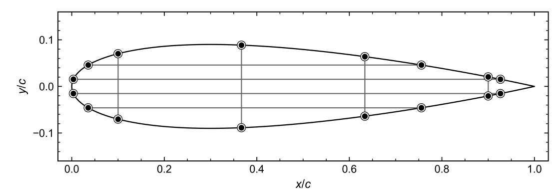
$$\boldsymbol{V} = \nabla\Phi. \tag{3}$$



(a) Horizontal model (M1)



(b) Vertical model (M2)



(c) Combined independent model (M3)

Figure 2: Porous-airfoil configurations considered in the present study: (a) horizontal model (M1), (b) vertical model (M2), and (c) combined independent model (M3).

Application of mass conservation gives Laplace's equation,

$$\nabla^2\Phi = 0. \tag{4}$$

The numerical solution is constructed by superposing the uniform freestream with constant-strength source distributions over the airfoil surface and a single constant vortex strength distributed around the complete airfoil boundary. The source distributions represent the airfoil thickness, while the global vortex strength introduces the circulation required to generate lift.

2.3.2. *Source–vortex panel formulation*

For an airfoil discretised into N panels, the aerodynamic unknowns are the N source strengths

$$\boldsymbol{\lambda} = \left[\lambda_1, \lambda_2, \dots, \lambda_N\right]^{\mathrm{T}} \tag{5}$$

and the global vortex strength γ.

The normal and tangential source influence coefficients are denoted by I_{ij} and J_{ij}, respectively, while K_{ij} and L_{ij} represent the corresponding vortex influence coefficients.

For an impermeable airfoil, the normal velocity at every control point is zero. To represent a porous opening, this condition is modified by prescribing a surface-normal transpiration velocity $v_{n,i}$,

$$\boldsymbol{V}_i \cdot \boldsymbol{n}_i = v_{n,i}. \tag{6}$$

The conventional impermeable boundary condition is recovered when

$$v_{n,i} = 0. \tag{7}$$

Following discretisation, the normal boundary condition for panel i becomes

$$\pi\lambda_i + \sum_{\substack{j=1\\ j\neq i}}^{N} I_{ij}\lambda_j - \gamma\sum_{j=1}^{N} K_{ij} = -2\pi V_\infty \cos\beta_i + 2\pi v_{n,i}, \quad (8)$$

where β_i denotes the angle between the local panel normal and the freestream direction.

The trailing-edge Kutta condition is imposed by requiring the tangential velocities on the first and final panels to be equal in magnitude and opposite in direction,

$$V_{t,1} + V_{t,N} = 0. \quad (9)$$

The N normal boundary conditions together with the Kutta condition produce an $(N+1)\times(N+1)$ linear system,

$$\begin{bmatrix} \boldsymbol{A}_\lambda & \boldsymbol{a}_\gamma \\ \boldsymbol{a}_K^{\mathrm{T}} & a_{K,\gamma} \end{bmatrix} \begin{bmatrix} \lambda \\ \gamma \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}(\boldsymbol{v}_n) \\ b_K \end{bmatrix}. \quad (10)$$

The influence matrix depends only on the discretised airfoil geometry. It is therefore assembled and LU-factorised once for each geometry, while the right-hand side is updated according to the freestream condition and surface-normal transpiration velocity.

2.3.3. Surface velocity and pressure

Following solution of the panel system, the tangential velocity on panel i is evaluated as

$$V_{t,i} = V_\infty \sin\beta_i + \frac{1}{2\pi}\sum_{j=1}^{N}\lambda_j J_{ij} + \frac{\gamma}{2} - \frac{\gamma}{2\pi}\sum_{j=1}^{N} L_{ij}. \quad (11)$$

At a porous opening, the local surface velocity contains both tangential and normal components. The velocity magnitude is therefore

$$V_i^2 = V_{t,i}^2 + v_{n,i}^2. \quad (12)$$

The pressure coefficient is obtained from the steady incompressible Bernoulli equation,

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + v_{n,i}^2}{V_\infty^2}. \quad (13)$$

The corresponding absolute pressure is

$$p_i = p_\infty + \frac{1}{2}\rho_\infty V_\infty^2 C_{p,i}. \quad (14)$$

2.4. Internal channel-flow model

The internal flow through each porous passage is modelled as pressure-driven flow through a circular channel. For channel k with diameter D_k, the cross-sectional area is

$$A_k = \frac{\pi D_k^2}{4}. \quad (15)$$

The mean internal velocity is related to the volumetric flow rate by

$$\overline{U}_k = \frac{Q_k}{A_k}, \quad (16)$$

where Q_k [$\mathrm{m^3\,s^{-1}}$] is the volumetric flow rate through the channel.

The internal flow is driven by the pressure difference between the two surface openings,

$$\Delta p_k = p_{1,k} - p_{2,k}. \quad (17)$$

The sign of Δp_k determines the direction of the internal flow. The corresponding channel Reynolds number is

$$Re_{D,k} = \frac{\rho_\infty|\overline{U}_k|D_k}{\mu_\infty}. \quad (18)$$

For fully developed laminar flow through a circular channel, the pressure-driven flow is described by the Hagen–Poiseuille equation,

$$Q_k = \frac{\pi D_k^4}{128\mu_\infty L_k}\Delta p_k. \quad (19)$$

This formulation naturally accounts for the direction of flow through the sign of Δp_k.

The pressure loss can equivalently be expressed using the Darcy–Weisbach relation,

$$|\Delta p_k| = f_k \frac{L_k}{D_k}\frac{\rho_\infty \overline{U}_k^2}{2}, \quad (20)$$

where f_k is the Darcy friction factor.

For fully developed laminar flow through a circular pipe,

$$f_{\mathrm{lam},k} = \frac{64}{Re_{D,k}}. \quad (21)$$

The Hagen–Poiseuille and Darcy–Weisbach formulations are therefore equivalent for fully developed laminar flow through the circular channels considered here.

2.5. Aero–hydraulic coupling

The aerodynamic and internal-flow models are coupled through the pressure and surface-normal velocity at the channel openings.

The pressure at each channel opening is obtained from the aerodynamic surface-pressure distribution. When an opening intersects multiple panels, its pressure is evaluated using a length-weighted average over the corresponding surface footprint. For opening m of channel k,

$$p_{m,k} = \frac{\sum_{i\in\mathcal{P}_{m,k}} p_i \ell_i}{\sum_{i\in\mathcal{P}_{m,k}} \ell_i}, \qquad m = 1, 2, \quad (22)$$

where $\mathcal{P}_{m,k}$ denotes the set of panels intersected by the opening and ℓ_i is the corresponding overlap length.

The two opening pressures provide the channel pressure difference,

$$\Delta p_k = p_{1,k} - p_{2,k}, \tag{23}$$

which is supplied to the internal-flow model to determine Q_k.

The mean velocity through each circular opening is

$$U_{\text{open},k} = \frac{Q_k}{A_k}. \tag{24}$$

This velocity is imposed on the external aerodynamic model as a surface-normal transpiration velocity. With the outward panel normal defined as positive, flow entering the airfoil is assigned a negative normal velocity, whereas flow leaving the airfoil is assigned a positive normal velocity,

$$v_{n,k}^{\text{in}} = -\frac{Q_k}{A_k}, \qquad v_{n,k}^{\text{out}} = +\frac{Q_k}{A_k}. \tag{25}$$

When an opening intersects multiple aerodynamic panels, the prescribed normal velocity is applied over the corresponding opening footprint so that the transpiration boundary condition is represented consistently by the discretised airfoil surface.

The updated surface-normal velocity modifies the panel-method boundary condition and therefore changes the external pressure distribution. The new opening pressures are subsequently supplied to the internal-flow model, and the aerodynamic and hydraulic solutions are coupled until the channel flow used to impose the opening velocity is consistent with that predicted from the resulting pressure difference.

A case is classified as converged when the prescribed coupling criterion is satisfied for all channels. Cases that do not satisfy the convergence requirement are reported as `NaN`.

2.6. Aerodynamic coefficients

The aerodynamic forces are obtained by integrating the pressure contribution over the discretised airfoil surface. The contribution of panel i to the global Cartesian force coefficients is

$$\mathrm{d}C_{x,i} = -C_{p,i} n_{x,i} \frac{S_i}{c}, \tag{26}$$

$$\mathrm{d}C_{y,i} = -C_{p,i} n_{y,i} \frac{S_i}{c}. \tag{27}$$

The total Cartesian force coefficients are

$$C_x = \sum_{i=1}^{N} \mathrm{d}C_{x,i}, \qquad C_y = \sum_{i=1}^{N} \mathrm{d}C_{y,i}. \tag{28}$$

Rotation into the freestream-aligned coordinate system gives the lift and pressure-drag coefficients,

$$C_L = C_y \cos\alpha - C_x \sin\alpha, \tag{29}$$

$$C_{D,p} = C_x \cos\alpha + C_y \sin\alpha. \tag{30}$$

The pitching-moment coefficient is evaluated about the quarter-chord reference location,

$$(x_{\text{ref}}, y_{\text{ref}}) = (0.25c, 0), \tag{31}$$

using

$$C_M = -\sum_{i=1}^{N} \frac{(x_i - x_{\text{ref}})\mathrm{d}C_{y,i} - (y_i - y_{\text{ref}})\mathrm{d}C_{x,i}}{c}. \tag{32}$$

For diagnostic comparison, an additional lift coefficient is calculated from the solved circulation using the Kutta–Joukowski relationship,

$$C_{L,\Gamma} = \frac{2\gamma \sum_{i=1}^{N} S_i}{V_\infty c}. \tag{33}$$

2.7. Numerical setup

The aerodynamic response of each porous configuration is evaluated relative to an impermeable reference airfoil using the same geometry, panel discretisation and operating conditions. An inviscid XFOIL solution may additionally be used for comparison of the impermeable-airfoil pressure distribution and angle-of-attack response.

Unless otherwise stated, the calculations are performed for a NACA 0018 airfoil with chord

$$c = 1 \text{ m}, \tag{34}$$

discretised using $N = 1000$ panels.

The freestream fluid properties are

$$\rho_\infty = 1.225 \text{ kg m}^{-3}, \tag{35}$$

$$\mu_\infty = 1.8 \times 10^{-5} \text{ Pa s}, \tag{36}$$

and

$$p_\infty = 101325 \text{ Pa}. \tag{37}$$

A chord Reynolds number of

$$Re_c = 5.0 \times 10^5 \tag{38}$$

is prescribed. The corresponding freestream velocity is calculated from

$$V_\infty = \frac{Re_c \mu_\infty}{\rho_\infty c}. \tag{39}$$

The principal fixed-angle analysis is performed at

$$\alpha = 4^\circ. \tag{40}$$

The angle-of-attack sweep extends from -5° to 15° in increments of 1°.

Circular channel diameters of

$$D = 8,\ 9,\ \text{and } 10 \text{ mm} \tag{41}$$

are considered.

Each angle of attack is solved independently, and the solution from the preceding angle is not used to initialise the subsequent case.

3. Results and discussion

The aerodynamic response of the three porous airfoil configurations was investigated over an angle-of-attack range of $-5°$ to $15°$. Three pore heights, $D = 8$, 9 and 10 mm, were considered for the horizontal (M1), vertical (M2) and combined horizontal–vertical (M3) configurations. The corresponding impermeable (solid airfoil) panel-method solution, and the inviscid XFOIL prediction are included as reference cases.

The influence of the porous channels is first assessed using the integrated lift and pitching-moment coefficients. The resulting aerodynamic trends are then related to changes in the external pressure and velocity fields. The flow-field comparison is presented for $\alpha = 10°$ and $D = 9$ mm.

3.1. Baseline aerodynamic validation

The impermeable source vortex panel solution is compared with the corresponding inviscid XFOIL prediction in Figs. 3 and 4. The two approaches show similar angle-of-attack trends for both lift and pitching moment over the investigated range.

The agreement between the two impermeable-airfoil solutions provides a reference for evaluating the aerodynamic changes introduced by the internal channels. Small differences between the panel-method and XFOIL predictions are expected because the two methods employ different numerical formulations, although both represent inviscid external flow.

The impermeable panel solution is therefore used as the primary reference when evaluating the effect of the porous configurations, ensuring that the porous and solid cases are compared using the same panel discretisation aerodynamic formulation.

Although the horizontal and vertical channels are physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field.

3.2. Effect of porous-channel configuration on lift

Figure 3 compares the variation of lift coefficient with angle of attack for the three porous configurations and channel diameters.

The horizontal configuration produces the largest change in lift. At positive angles of attack, M1 develops a higher lift coefficient than the impermeable reference, with the difference increasing as the angle of attack increases. The effect also becomes stronger with channel diameter, with the $D = 10$ mm configuration producing the highest lift at larger positive angles of attack. A corresponding increase in the magnitude of negative lift is observed at negative angles of attack, giving an approximately antisymmetric response consistent with the symmetric NACA 0018 geometry.

For M1, both openings of each channel are located on the same airfoil surface. The streamwise pressure variation therefore generates a pressure difference between the two openings and drives internal flow from the locally higher-pressure location towards the lower-pressure location. Suction at one opening and blowing at the other redistribute the external velocity and pressure along the surface. At positive angle of attack, the different pressure distributions on the upper and lower surfaces cause this redistribution to produce a net increase in the aerodynamic loading and hence lift.

The stronger diameter dependence of M1 is consistent with the reduction in hydraulic resistance as the channel diameter increases. A larger diameter permits a greater internal flow for a given pressure difference, strengthening the interaction between the channel flow and the external aerodynamic field.

The vertical configuration exhibits a distinctly different behaviour. At positive angles of attack, the lift coefficient is generally lower than that of the impermeable airfoil, while the three diameter cases remain relatively close over most of the angle-of-attack range.

In M2, each channel connects the higher-pressure lower surface to the lower-pressure upper surface. The resulting pressure difference drives flow from the pressure surface towards the suction surface. This cross-surface transfer partially reduces the pressure difference responsible for lift, which is consistent with the lower lift coefficient obtained for the vertical configuration. The relatively small separation between the diameter cases indicates that M2 is less sensitive to channel diameter than M1.

The combined horizontal–vertical configuration remains comparatively close to the impermeable reference. At positive angles of attack, its lift coefficient is generally slightly lower than the solid-airfoil solution, and the three diameter cases remain closely grouped.

This response reflects the simultaneous action of the two channel arrangements. The horizontal passages redistribute flow between different chordwise locations on the same surface, whereas the vertical passages transfer flow between the pressure and suction surfaces. Consequently, the lift-enhancing redistribution associated with the horizontal passages is partly offset by the pressure equalisation introduced by the vertical passages.

The horizontal and vertical channels in M3 are physically separate and hydraulically independent; however, their aerodynamic effects interact through the external pressure field. Changes produced by one set of channels alter the surface pressures that drive the other set, so the combined response is not a simple superposition of M1 and M2.

Overall, the lift results demonstrate that channel orientation and the pressure difference between the connected surface locations have a stronger influence on aerodynamic performance than channel diameter alone. M1 produces the largest lift enhancement and the strongest diameter dependence, whereas M2 and M3 exhibit considerably weaker sensitivity.

3.3. Effect of porous-channel configuration on pitching moment

The corresponding quarter-chord pitching-moment coefficients are presented in Fig. 4.

The horizontal configuration produces the largest departure from the impermeable-airfoil response. As the positive

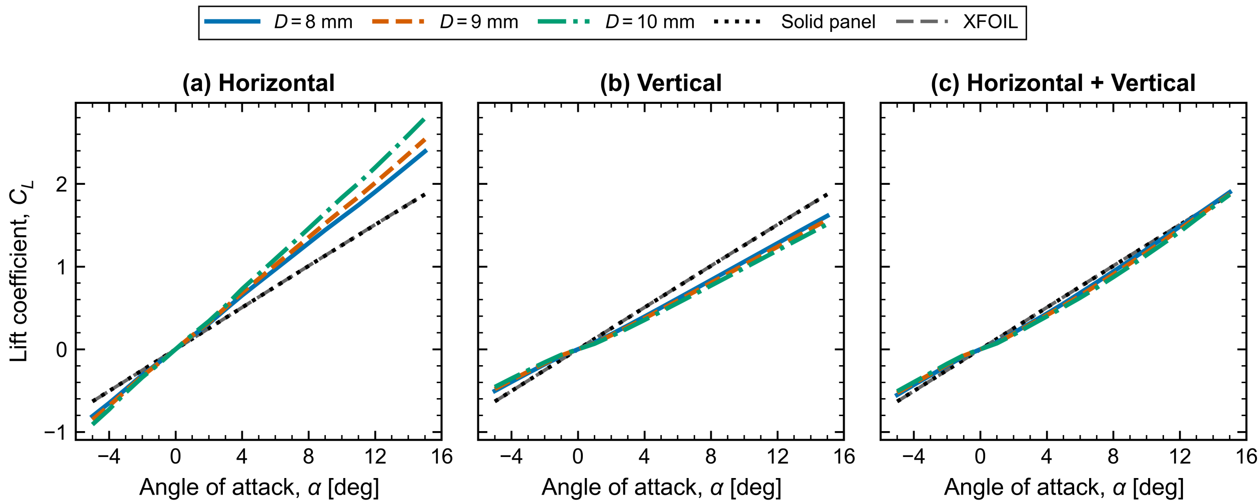


Figure 3: Lift coefficient as a function of angle of attack for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Results are shown for circular-channel diameters $D = 8$, 9 and 10 mm together with the impermeable panel-method and XFOIL reference solutions.

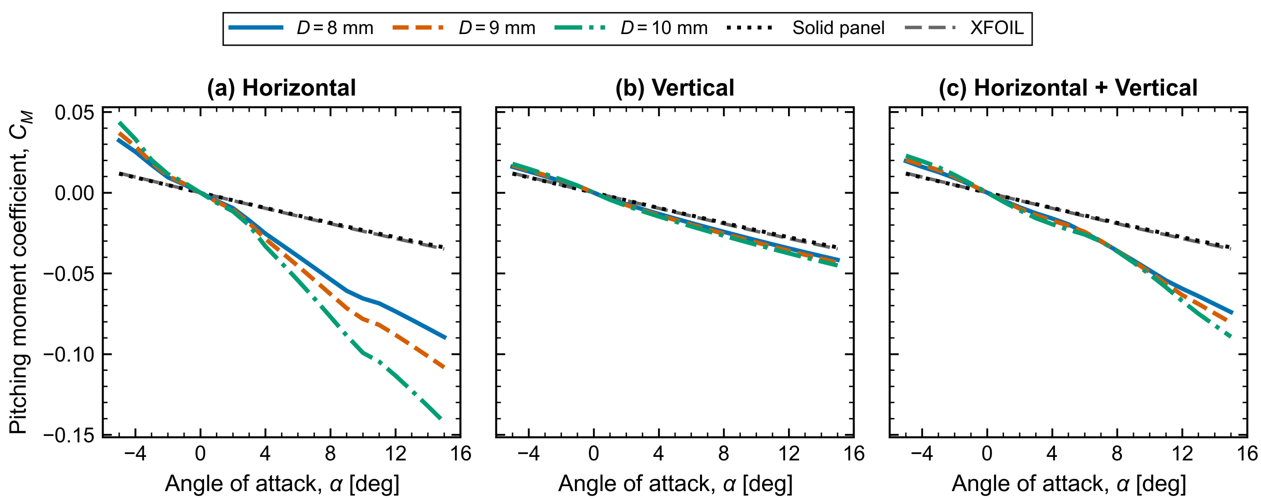


Figure 4: Pitching-moment coefficient about the quarter chord as a function of angle of attack for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Results are shown for circular-channel diameters $D = 8$, 9 and 10 mm together with the impermeable panel-method and XFOIL reference solutions.

angle of attack increases, the pitching-moment coefficient becomes progressively more negative, with the magnitude of the change increasing with channel diameter. The $D =$ 10 mm configuration therefore produces the largest negative pitching moment at higher positive angles of attack.

This behaviour reflects the chordwise redistribution of aerodynamic loading produced by M1. Because the two openings of each horizontal channel are located at different chordwise positions, the pressure changes associated with suction and blowing act at different distances from the quarter-chord reference point. The resulting changes in local loading therefore generate an additional pitching moment. Increasing the channel diameter strengthens the internal flow and consequently increases this chordwise redistribution.

The lift enhancement produced by M1 is therefore accompanied by a larger negative pitching moment. This represents an important aerodynamic trade-off, since the configuration that provides the largest increase in lift also produces the greatest change in the longitudinal moment characteristics.

The vertical configuration produces substantially smaller changes in pitching moment, and the three diameter cases remain closely grouped throughout most of the angle-of-attack range. This is consistent with the weaker diameter dependence observed in the lift results.

For M2, the upper and lower openings of each vertical channel are located at approximately the same chordwise position. The resulting pressure redistribution therefore occurs predominantly between the two airfoil surfaces rather than between widely separated chordwise locations. Consequently, the associated change in moment arm is smaller than for M1, resulting in a weaker pitching-moment response.

The combined configuration exhibits an intermediate behaviour. The pitching moment becomes increasingly negative with positive angle of attack, although the departure from the impermeable solution remains smaller than for M1. The influence of channel diameter also becomes more apparent at higher positive angles of attack.

In M3, the horizontal passages redistribute loading between different chordwise positions, while the vertical passages primarily redistribute pressure between the upper and lower surfaces. Although the two channel systems remain physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field. The resulting pitching-moment behaviour therefore reflects the combined influence of these two loading mechanisms.

Overall, the pitching-moment results demonstrate that the aerodynamic effect of the porous channels depends not only on the magnitude of the pressure redistribution but also on its chordwise location. M1 produces the largest moment change because its passages redistribute loading between spatially separated chordwise positions, whereas M2 produces a smaller response because its pressure redistribution occurs mainly between the upper and lower surfaces at similar chordwise locations.

3.4. Flow-field characteristics of porous airfoils

To examine the mechanisms responsible for the changes in the integrated aerodynamic coefficients, the pressure and velocity fields of the porous airfoils are compared with the corresponding impermeable-airfoil solution at

$$\alpha = 10^\circ, \qquad D = 9\ \text{mm}. \tag{42}$$

This condition represents a positive angle of attack at which the aerodynamic influence of the porous channels is clearly developed.

3.4.1. Pressure-field characteristics

Figure 5 shows the pressure difference

$$\Delta p = p_{\text{porous}} - p_{\text{solid}}, \tag{43}$$

for the three porous configurations.

The difference field is used instead of the absolute pressure distribution to isolate the aerodynamic changes introduced by the porous channels. In the absolute field, the dominant pressure gradients associated with the baseline airfoil, particularly near the leading edge, can obscure the comparatively local influence of the channel flow. The difference contours therefore provide a clearer indication of the location, magnitude and sign of the pressure redistribution.

The strongest pressure differences occur near the channel openings, where neighbouring regions of positive and negative Δp form a dipole-like pressure signature. This behaviour results from the local suction and blowing associated with the two ends of a pressure-driven channel, which perturb the external flow in opposite directions.

The aerodynamic significance of Δp depends on the surface on which it occurs. At positive angle of attack, a negative Δp on the upper suction surface increases the suction and therefore tends to increase lift, whereas a positive Δp reduces the suction and tends to decrease lift. On the lower pressure surface, the opposite applies: a positive Δp contributes positively to lift, while a negative value reduces the pressure loading.

For M1, the pressure changes remain predominantly localised around the chordwise channel openings. Since each passage connects two locations on the same surface, the paired positive and negative regions represent a redistribution of pressure along the chord. Although these disturbances are local, their combined effect increases the pressure difference between the upper and lower surfaces, consistent with the larger lift coefficient obtained for the horizontal configuration.

For M2, each channel directly connects the pressure and suction surfaces. The resulting internal flow transfers fluid from the higher-pressure lower surface towards the lower-pressure upper surface. This produces local pressure disturbances at the openings while also partially equalising the pressure between the two surfaces. The resulting pressure distribution is therefore consistent with the lower lift coefficient observed for M2.

The combined configuration exhibits characteristics of both mechanisms. The horizontal channels redistribute pressure along the same surface, whereas the vertical channels transfer flow between the pressure and suction surfaces. Although the two channel systems are physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field. Consequently, the pressure equalisation associated with the vertical channels partly offsets the lift-enhancing redistribution produced by the horizontal channels, giving an overall response closer to the impermeable airfoil.

3.4.2. Velocity-field characteristics

The velocity field is similarly presented as the difference between the porous and solid-airfoil solutions,

$$\Delta|\boldsymbol{V}| = |\boldsymbol{V}|_{\text{porous}} - |\boldsymbol{V}|_{\text{solid}}. \tag{44}$$

This representation removes the dominant background acceleration around the airfoil and highlights the local acceleration and deceleration generated by the channel flow. The largest velocity differences again occur near the channel openings, where suction and blowing introduce a surface-normal velocity into the external flow. Neighbouring regions of increased and reduced velocity form an acceleration–deceleration pattern that corresponds closely to the dipole-like pressure features in Fig. 5.

The relationship between the pressure and velocity contours is consistent with Bernoulli's equation: a local increase

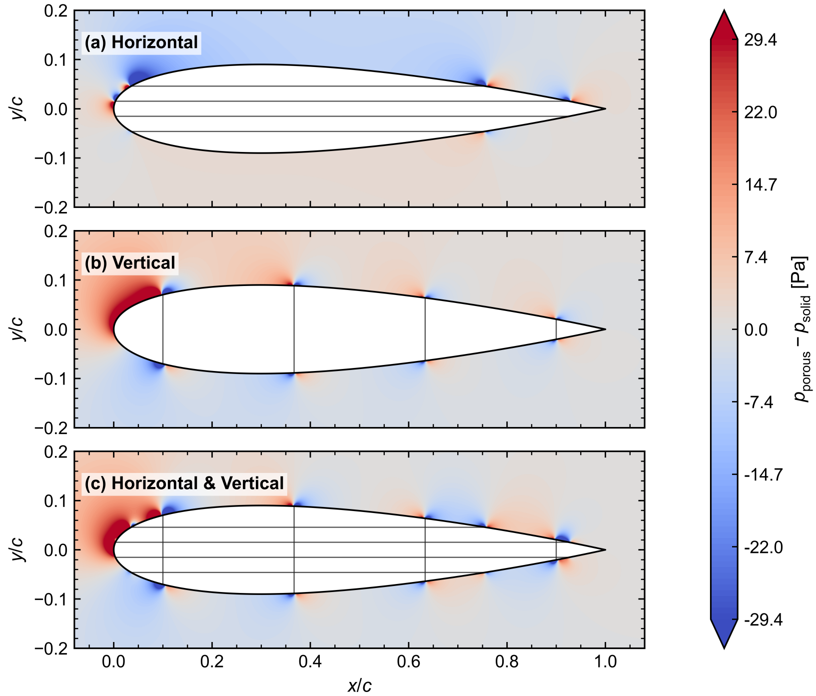


Figure 5: Pressure-field difference between the porous and impermeable airfoils at $\alpha = 10°$ and $D = 9$ mm for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Positive values indicate an increase in pressure relative to the impermeable airfoil, while negative values indicate a reduction.

in velocity is generally associated with a decrease in static pressure, whereas a local reduction in velocity corresponds to an increase in pressure. The velocity contours therefore provide a complementary view of the flow mechanism responsible for the pressure redistribution.

For M1, alternating regions of acceleration and deceleration occur around the chordwise openings. Flow entering one end of a passage produces a local suction effect, while discharge from the other end produces a local blowing effect. These changes alter the surface velocity distribution on both the suction and pressure surfaces. Their combined effect strengthens the pressure difference across the airfoil, consistent with the increased lift and larger pitching-moment change obtained for the horizontal configuration.

For M2, the pressure difference between the lower and upper surfaces drives flow from the pressure surface towards the suction surface. Suction at the lower opening and blowing at the upper opening alter the velocity on both surfaces and reduce part of the velocity contrast between them. This is consistent with the partial pressure equalisation and the corresponding reduction in lift. A broader velocity-field difference is also visible near the leading edge, where the baseline pressure and velocity gradients are strongest.

The combined configuration contains the local acceleration–deceleration features of M1 together with the upper-to-lower surface interaction associated with M2. Although the horizontal and vertical channels remain hydraulically independent, their aerodynamic effects interact through the external flow field. The resulting velocity distribution is therefore not a simple superposition of the two individual configurations and is consistent with the M3 lift coefficient remaining closer to that of the impermeable airfoil.

Overall, the difference contours demonstrate that the aerodynamic influence of the porous channels is concentrated primarily around the channel openings and depends strongly on channel orientation. The horizontal passages redistribute the external flow predominantly along the same surface, whereas the vertical passages transfer flow between the pressure and suction surfaces. These different mechanisms account for the contrasting lift and pitching-moment behaviour of M1, M2 and M3.

3.5. Overall comparison of porous configurations

The combined aerodynamic and flow-field results show that the response of the porous airfoil is governed primarily by the orientation and placement of the internal channels, rather than by channel diameter alone.

Among the configurations investigated, the horizontal model (M1) produces the strongest aerodynamic effect. At positive angles of attack, M1 increases the lift coefficient relative to the impermeable airfoil, with the enhancement becoming more pronounced as the channel diameter increases. This increase in lift is accompanied by a more negative pitching moment, particularly for the larger channel diameters, indicating a significant redistribution of the aerodynamic loading along the chord.

The vertical model (M2) exhibits a distinctly different behaviour. The channels transfer flow between the higher-pressure lower surface and the lower-pressure upper surface, promoting partial pressure equalisation across the airfoil.

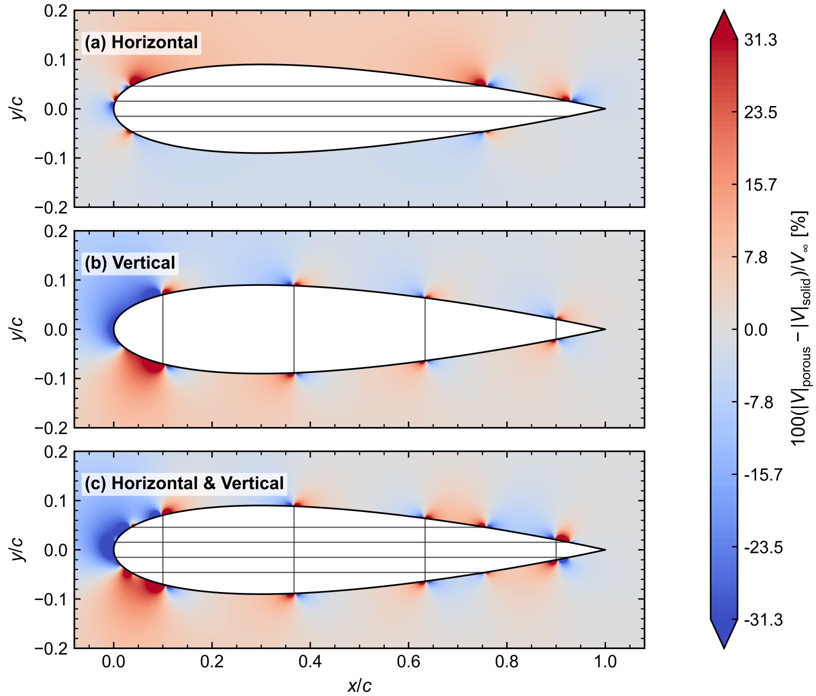


Figure 6: Difference in velocity magnitude between the porous and impermeable-airfoil solutions at $\alpha = 10°$ and $D = 9$ mm for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Positive values indicate an increase in velocity magnitude, while negative values indicate a reduction relative to the impermeable solution.

Consequently, the lift coefficient is generally reduced relative to the impermeable reference at positive angles of attack. The comparatively small separation between the three diameter cases also indicates a weaker sensitivity to channel diameter, with correspondingly smaller changes in pitching moment.

The combined horizontal–vertical model (M3) exhibits characteristics of both configurations but does not represent a simple superposition of their individual aerodynamic effects. The horizontal and vertical channels are physically separate and hydraulically independent; however, they interact indirectly through the external pressure and velocity fields. The pressure redistribution generated by the vertical passages partly offsets the lift-enhancing effect of the horizontal passages, resulting in an aerodynamic response that remains closer to the impermeable-airfoil solution.

The pressure- and velocity-difference fields provide further support for these trends. The strongest local changes occur near the channel openings, where suction and blowing alter the surrounding external flow. For M1, these effects primarily redistribute the flow between different chordwise locations on the same surface, whereas M2 transfers flow between the pressure and suction surfaces. The resulting aerodynamic behaviour therefore depends strongly on which surface regions are connected and on the pressure difference established between the channel openings.

Overall, M1 provides the greatest potential for lift enhancement among the three configurations considered. However, its larger change in pitching moment demonstrates that porous-channel performance should not be assessed from lift alone. The orientation, placement and diameter of the channels must instead be considered together with their effects on both aerodynamic forces and moments when selecting an appropriate porous-airfoil configuration.

4. Conclusion